

Cardinality and Finiteness

- $|A|$ (read “the cardinality of set A ”) is a measure of how many different elements A has. (**Same as $n(A)$**)

- E.g.,

$$|\emptyset| = 0,$$

$$|\{1,2,3\}| = 3,$$

$$|\{a,b\}| = 2,$$

$$|\{\{1,2,3\},\{4,5\}\}| = 2$$

- We say A is infinite if it is not finite.
- What are some infinite sets we’ve seen?

N Z R

The Power Set Operation

- The power set $P(A)$ of set A is the set of all subsets of A .

$$P(A) = \{x \mid x \subseteq A\}.$$

- E.g. If $A = \{a, b\}$ then $P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

- **Cardinality of power set - $P(A)$ is given by:**

$$|P(A)| = 2^{|A|} \quad (\text{for finite } A)$$

- It turns out that $|P(A)| > |A|$.

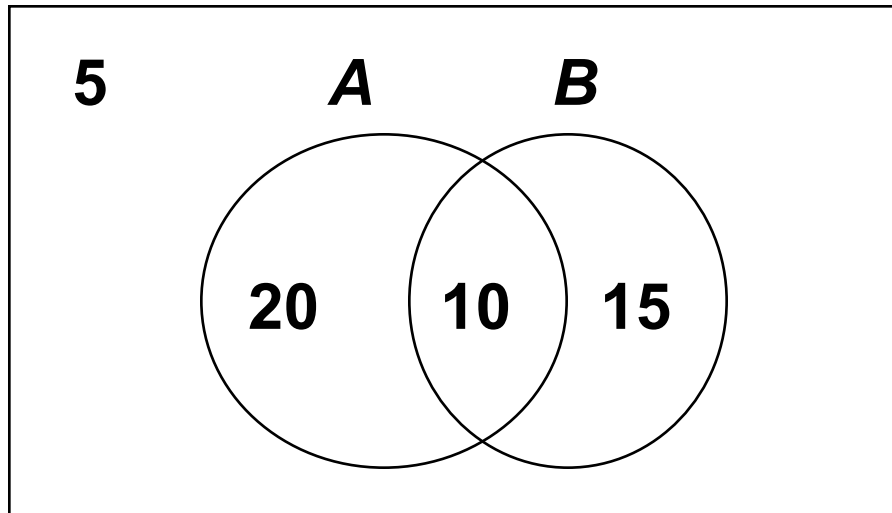
Question

1. If $A=\{a,b,c\}$, how many subsets can be formed ? What are they ?
2. How many proper subsets exist? What are they?
3. Identify the $P(A)$ (or power set A)?
4. What is the cardinality of A ?
5. Identify the cardinality of $P(A)$ (or power set A)

Counting and Venn Diagrams

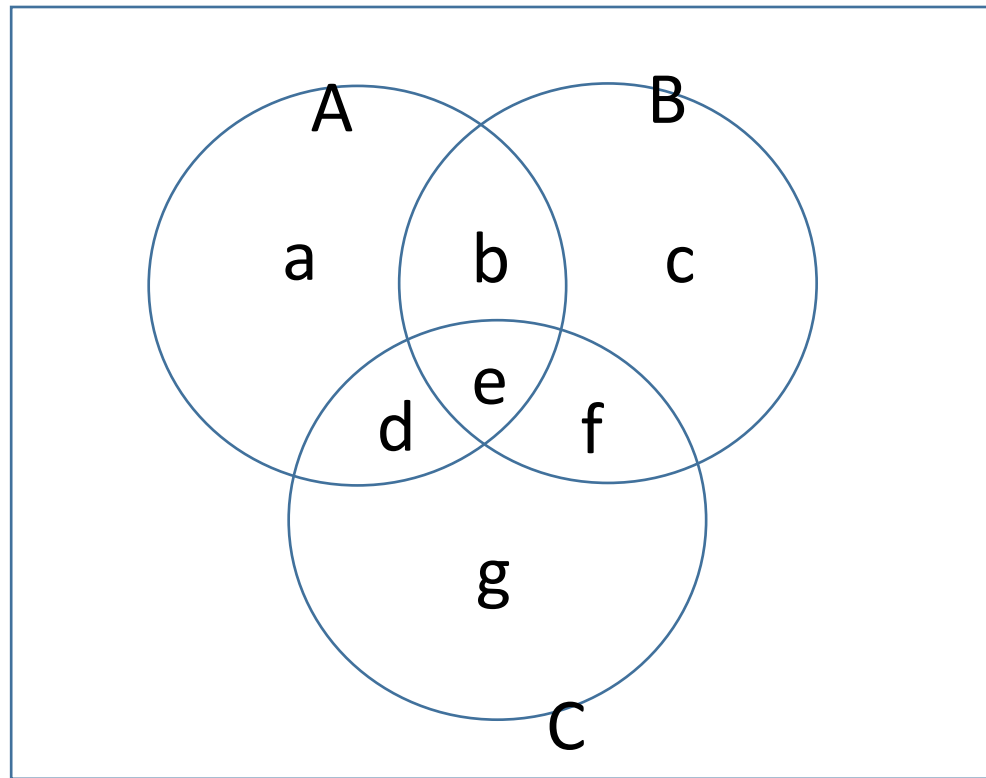
Example 1: In a class of 50 college freshmen, 30 are studying BASIC, 25 are studying PASCAL, and 10 are studying both. How many freshmen are studying either computer language?

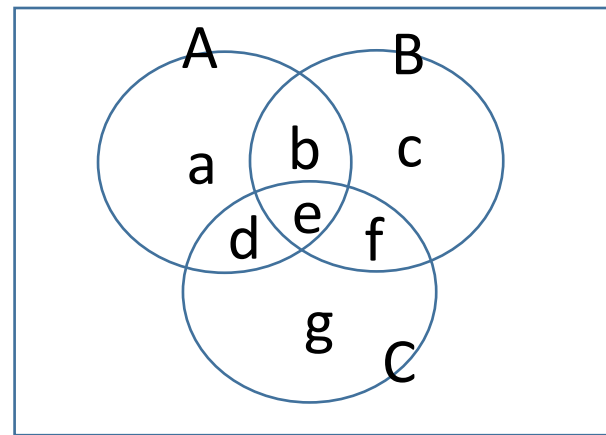
U



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Counting and Venn Diagrams (Con.)





- $|A| = a + b + d + e$

- $|B| = b + c + e + f$

- $|C| = d + e + f + g$

$$|A| + |B| + |C| = a + c + g + 2b + 2d + 2f + 3e$$

- $|A \cap B| = b + e$

—

- $|B \cap C| = e + f$

- $|A \cap C| = d + e$

$$|A \cap B| + |B \cap C| + |A \cap C| = b + d + f + 3e$$

+

- $|A \cap B \cap C| = e$

$$|A \cap B \cap C| = e$$

$$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C| = a + b + c + d + e + f + g$$

$$= |A \cup B \cup C|$$

Example 2: Defect types of a product:

D1: Level 1 Defect

D2: Level 2 Defect

D3: Level 3 Defect

Given 100 samples

set A: with D1

set B: with D2

set C: with D3

with $|A|=23$, $|B|=26$, $|C|=30$,

$|A \cap B|=7$, $|A \cap C|=8$, $|B \cap C|=10$,

$|A \cap B \cap C|=3$, how many samples have defects?

Counting and Venn Diagrams (Con.)

Given 100 samples

set A: with D1

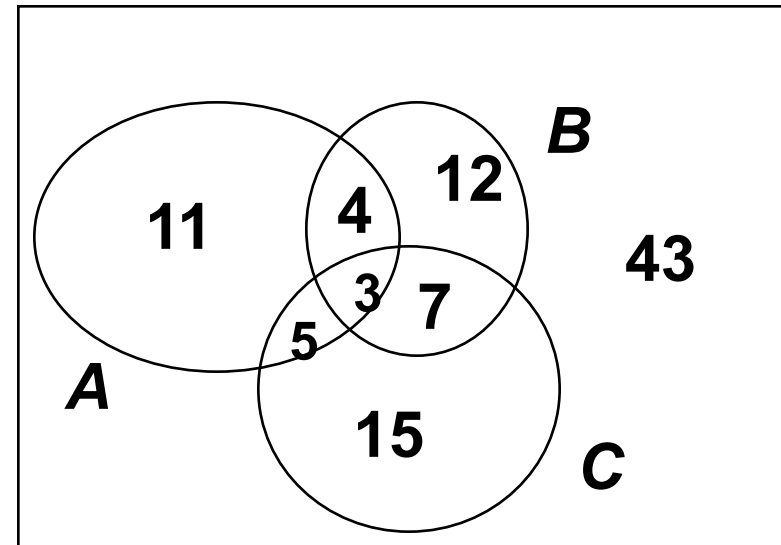
set B: with D2

set C: with D3

with $|A|=23$, $|B|=26$, $|C|=30$,

$|A \cap B|=7$, $|A \cap C|=8$, $|B \cap C|=10$,

$|A \cap B \cap C|=3$



$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

how many samples have defects?

Ans:57

Questions on Venn Diagrams

1.1 In a class there are 30 students. 21 students like Mathematics, 16 like English, 6 students do not like Mathematics or English. How many Students like both Mathematics and English?

1.2 In a class there are;

8 students who play football and hockey, 7 students who do not play football or hockey, 13 students who play hockey, 19 students who play football.

How many student are there in the class?

Question

1.3

The following data refers to random sample of 200 attending a seminar; they are categorized according to their education qualifications.

- There are 115 BSc Engineers, 100 CIMA qualified people and 70 AAT qualified people. 40 are both BSc and CIMA qualified. 50 are both CIMA and AAT qualified. 20 BSC Engineers are AAT qualified. 10 have no qualification.
- Required: Find the no of people who have all three qualifications?

Laws of the Algebra of Sets

- Idempotent laws

$$A \cup A = A$$

$$A \cap A = A$$

- Associative laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

- Commutative laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Laws of the Algebra of Sets

- Distributive laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- Identity laws

$$A \cup \emptyset = A \qquad A \cap \Omega = A$$

$$A \cup \Omega = \Omega \qquad A \cap \emptyset = \emptyset$$

- Involution law

$$\bar{\bar{A}} = A \quad \text{or} \quad A'' = A$$

Laws of the Algebra of Sets

- Complement laws

$$A \cup A' = \Omega$$

$$A \cap A' = \emptyset$$

$$\Omega' = \emptyset$$

$$\emptyset' = \Omega$$

- De Morgan's laws

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Relations & Functions

Outline:

- Cartesian Product
- Ordered pairs
- Arrow Diagrams
- Directed Graphs
- Functions

Cartesian Product

Definition : Given two non-empty sets A and B, the set of all ordered pairs (x, y) , where $x \in A$ and $y \in B$ is called Cartesian product of A and B

symbolically, we write $A \times B$

If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$

then $A \times B = \{(1, 4), (2, 4), (3, 4), (1, 5), (2, 5), (3, 5)\}$

and $B \times A = \{(4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3)\}$

Cartesian Products of Sets

- For sets A , B , their Cartesian product
 $A \times B \equiv \{(a, b) \mid a \in A \wedge b \in B\}$.
- E.g. $\{a, b\} \times \{1, 2\} = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$

Find the following Cartesian Products of sets

Q1) Let $A=\{1,2\}$ and $B=\{a,b,c\}$

- i. Find $A \times B$
- ii. $B \times A$?
- iii. $A \times A$?

Note:

- The Cartesian product deals with ordered pairs, so the order in which the sets are considered is important
- For finite A, B $n(A \times B) = n(A) \cdot n(B)$

Find the following Cartesian Products of sets

Q2) If $A=\{a,b\}$ and $B=\{1,2,3\}$, Find $A \times B$

Q3) If $A=\{a,b\}$, $B=\{1,2\}$, $C=\{2,3\}$ Find

- i. $A \times (B \cup C)$
- ii. $(A \times B) \cup (A \times C)$
- iii. $A \times (B \cap C)$
- iv. $(A \times B) \cap (A \times C)$

Relations

- A “relation” is just a relationship between sets of information.
- Think of all the students in your class and think of their heights. The pairing of names and heights is a relation.
- In relations, the pairs of names and heights are “ordered”, which means one comes first and the other comes second.

Relations - Definition

A Relation **R** from a non-empty set A to a non-empty set B is a **subset of the Cartesian product set $A \times B$** .

The subset is derived by describing a relationship between the first element and the second element of the ordered pairs in $A \times B$.

The set of all first elements in a relation R, is called the **domain** of the relation R, and the set of all second elements, is called the **range** of R.

Definition (Cont.)

- Let A and B be sets. A binary relation or simply a relation from A to B is a subset of $A \times B$.

Suppose R is a relation from A to B . Then R is the set of ordered pairs where each first element comes from A and each second element comes from B .

Relations

Ordered Pairs

Relations will be defined in terms of ordered pairs (a,b) of elements, where a is designated as the first element and b as the second element.

$$(a,b) = (c,d) \text{ if and only if } a = c \text{ and } b = d$$

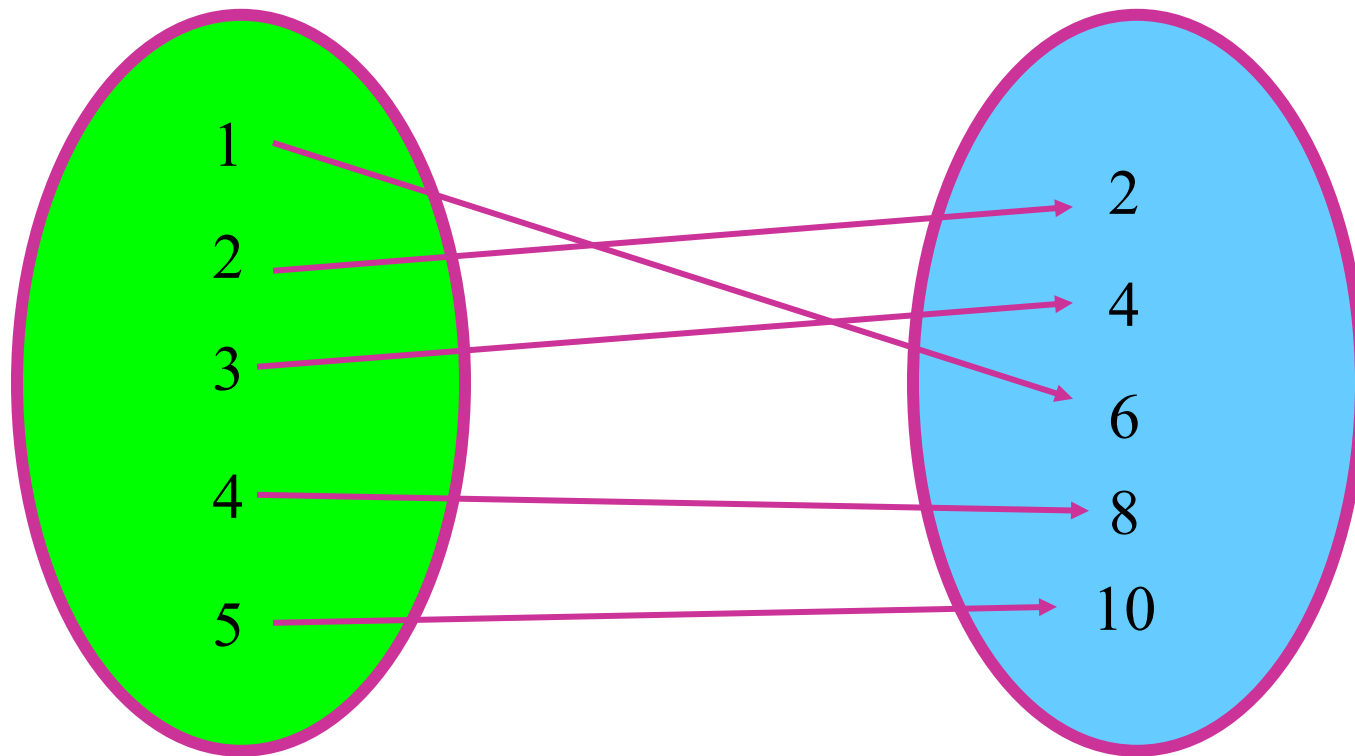
Pictorial representations of relations

- Now we look at different ways of picturing and representing binary relations.

1. Arrow Diagram:

Suppose A and B are two finite sets. Write down the elements of A and the elements of B in two disjoint disks. Then draw an arrow from $a \in A$, to $b \in B$ whenever a is related to b . This picture is called the arrow diagram of the relation

Pictorial representations of relations



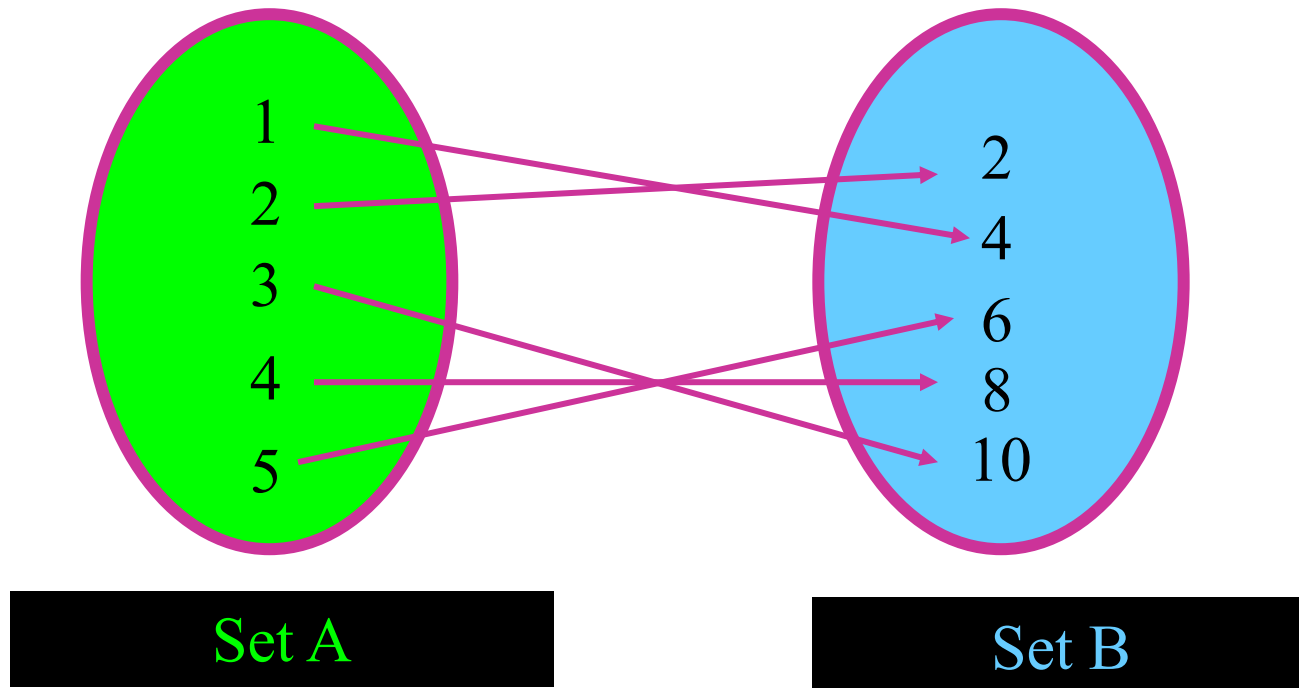
Domain (set of all x's)

Range (set of all y's)

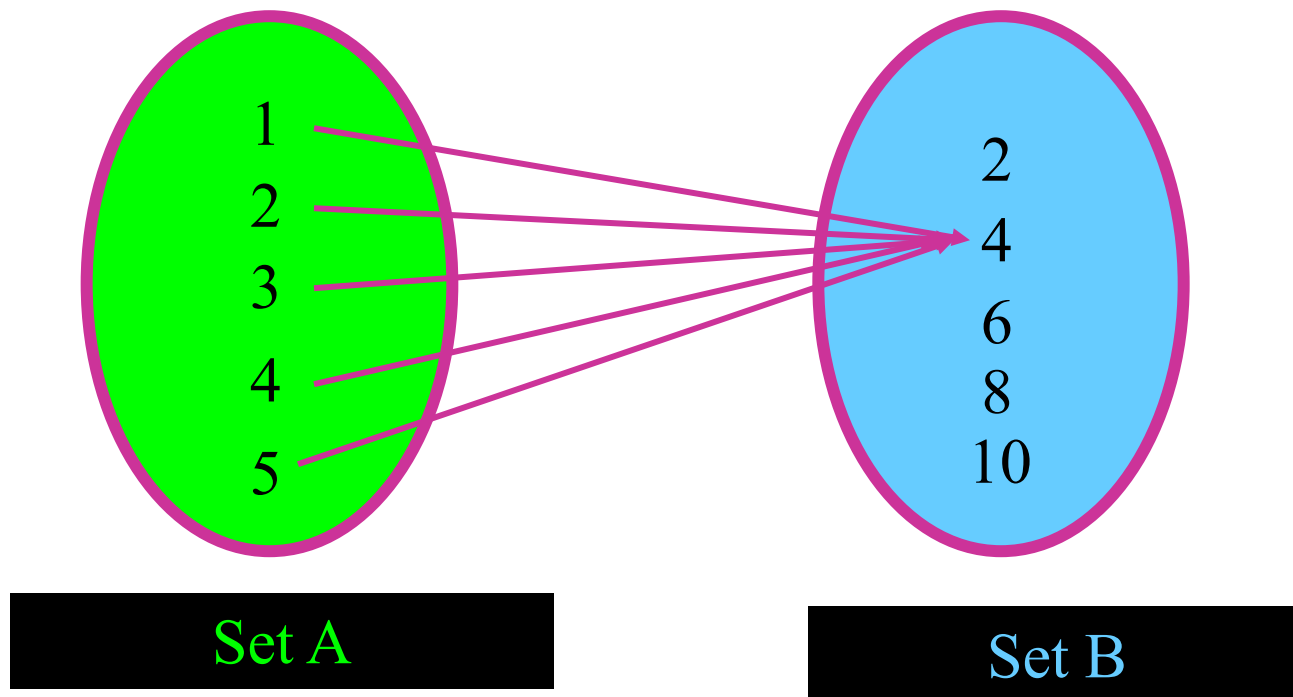
This relation can be written $\{(1,6), (2,2), (3,4), (4,8), (5,10)\}$

Q) Write the following relations in ordered pairs

1.



2.



Pictorial representations of relations

2. Directed Graphs

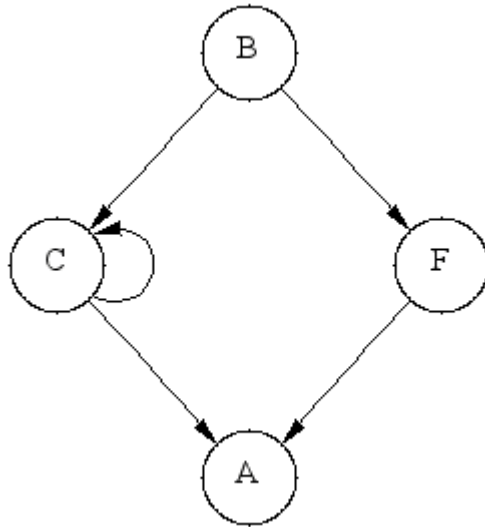
These graphs are used to picture a relation R , which is a relation from a finite set to itself.

Draw the directed graph of the relation R on the set $A = \{1, 2, 3, 4\}$,

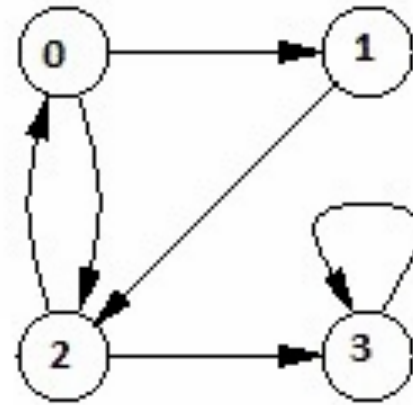
Let $R = \{(1,2), (2,2), (2,4), (3,2), (3,4), (4,1), (4,3)\}$

Find the relations of the following directed graphs

1.

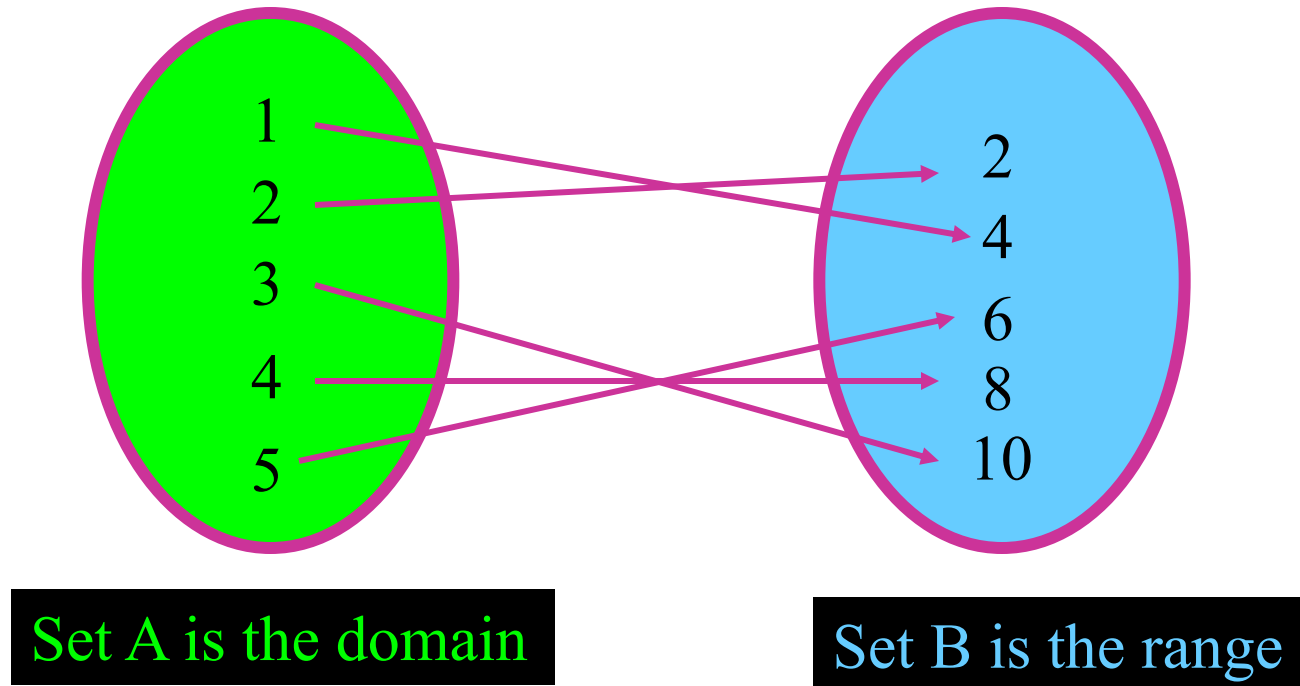


2.



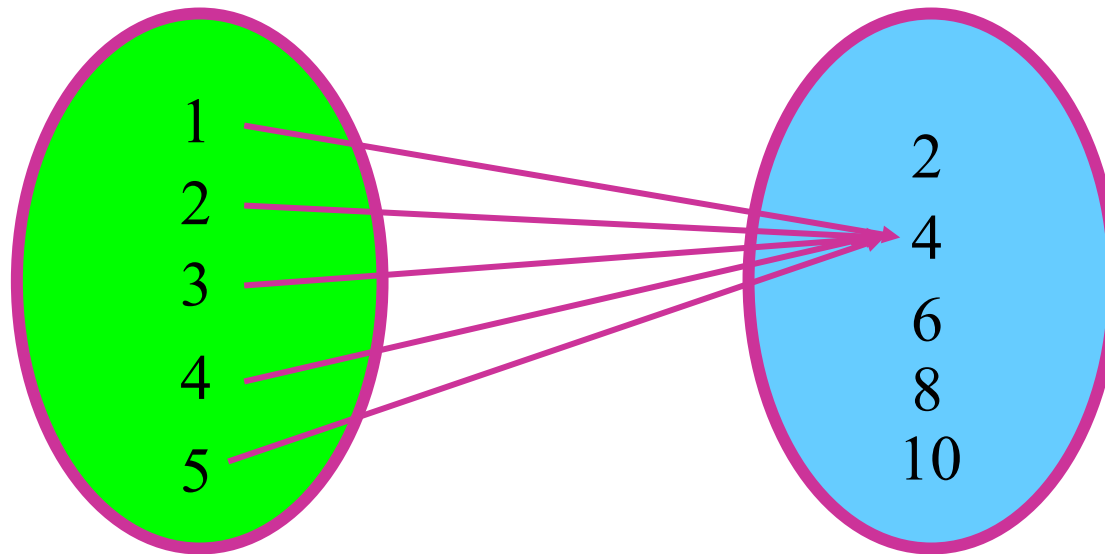
Functions

- A function f is a **relation** in which each element of the domain is paired with exactly one element of the range.



Functions

Is the following relation a function?



Set A is the domain

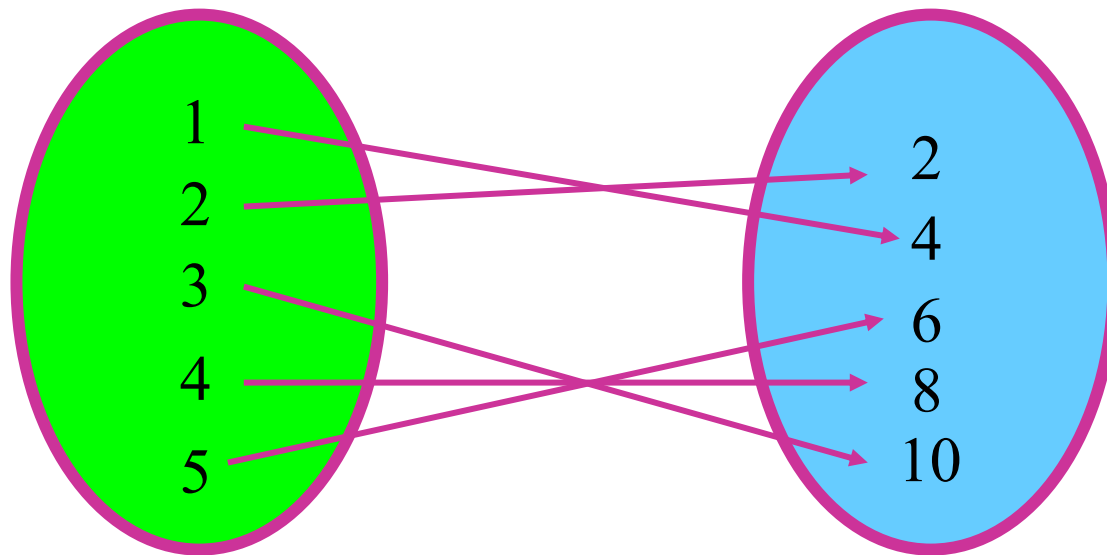
Set B is the range

Must use all the x's

The x value can only be assigned to one y

Functions

Is the following relation a function?

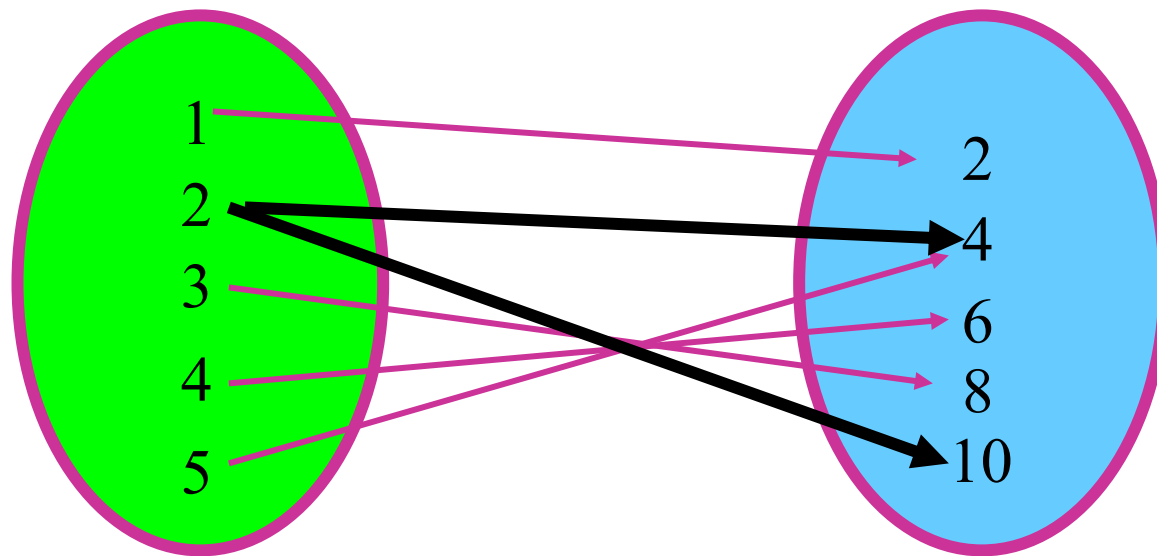


Set A is the domain

Set B is the range

Functions

Is the following relation a function?



Set A is the domain

Set B is the range